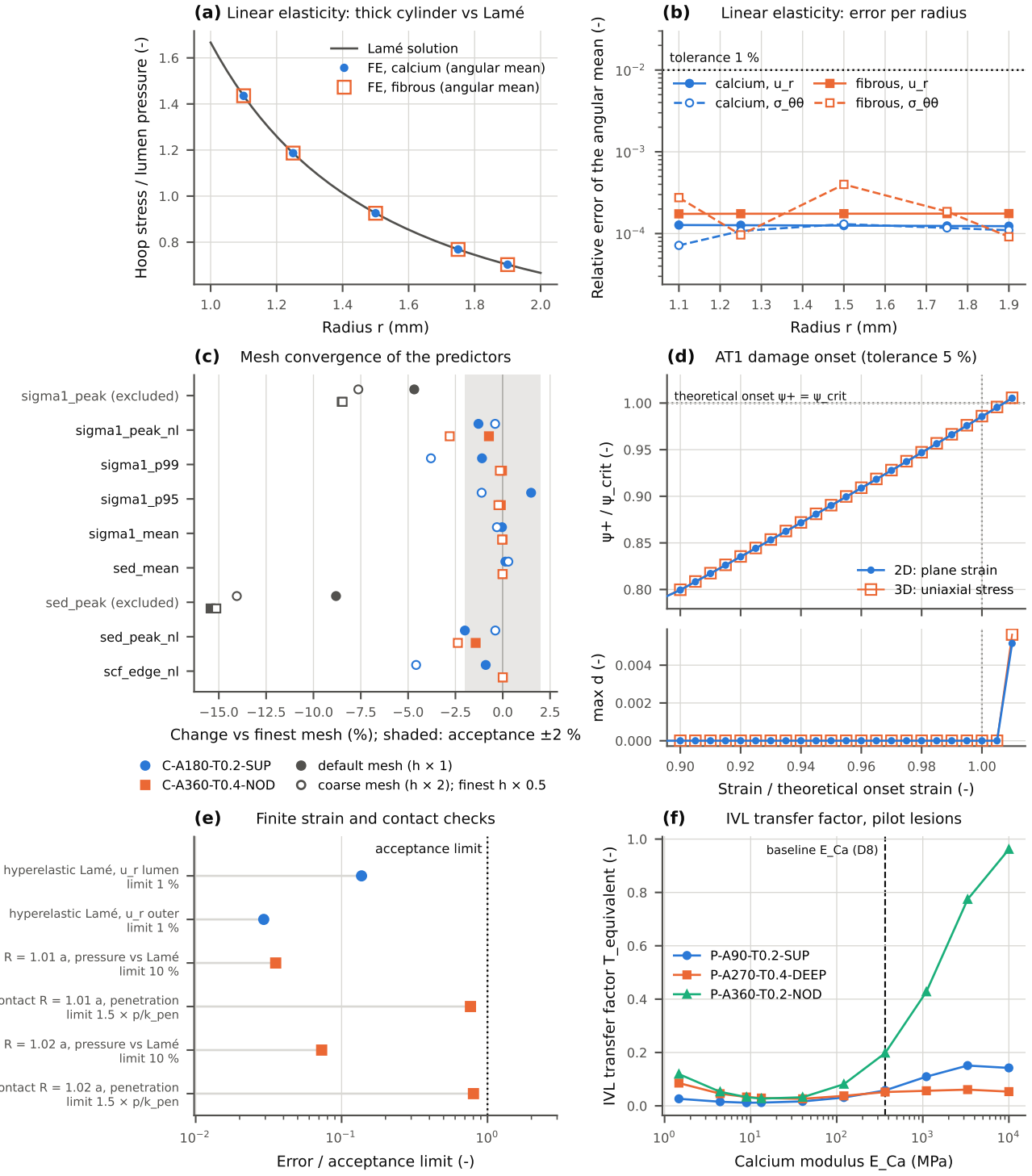
(a) Linear elasticity: thick cylinder vs Lamé
Lamé solution
FE, calcium (angular mean)
FE, fibrous (angular mean)
Hoop stress / lumen pressure (-)
Radius r (mm)
(b) Linear elasticity: error per radius
tolerance 1 %
calcium, u_r
fibrous, u_r
calcium, σ_θθ
fibrous, σ_θθ
Relative error of the angular mean (-)
Radius r (mm)
(c) Mesh convergence of the predictors
sigma1_peak (excluded)
sigma1_peak_nl
sigma1_p99
sigma1_p95
sigma1_mean
sed_mean
sed_peak (excluded)
sed_peak_nl
scf_edge_nl
Change vs finest mesh (%); shaded: acceptance ±2 %
C-A180-T0.2-SUP
C-A360-T0.4-NOD
default mesh (h × 1)
coarse mesh (h × 2); finest h × 0.5
(d) AT1 damage onset (tolerance 5 %)
theoretical onset ψ+ = ψ_crit
ψ+ / ψ_crit (-)
2D: plane strain
3D: uniaxial stress
max d (-)
Strain / theoretical onset strain (-)
(e) Finite strain and contact checks
acceptance limit
hyperelastic Lamé, u_r lumen
limit 1 %
hyperelastic Lamé, u_r outer
limit 1 %
contact R = 1.01 a, pressure vs Lamé
limit 10 %
contact R = 1.01 a, penetration
limit 1.5 × p/k_pen
contact R = 1.02 a, pressure vs Lamé
limit 10 %
contact R = 1.02 a, penetration
limit 1.5 × p/k_pen
Error / acceptance limit (-)
(f) IVL transfer factor, pilot lesions
baseline E_Ca (D8)
P-A90-T0.2-SUP
P-A270-T0.4-DEEP
P-A360-T0.2-NOD
IVL transfer factor T_equivalent (-)
Calcium modulus E_Ca (MPa)